

■ B.1.5 Special Objects

There are a number of types of special objects, which form the indivisible elements of expressions in *Mathematica*.

These objects have heads which are symbols that can be thought of as “tagging” their types. The objects contain “raw data”, which can usually be accessed only by functions specific to the particular type of object. You can extract the head of a special object using `Head`, but you cannot explicitly extract any of its other parts.

<code>Symbol</code>	symbol
<code>String</code>	character string <code>"cccc"</code>
<code>Integer</code>	integer <code>nnnn</code>
<code>Real</code>	real (floating point) number <code>nnn.nnn</code>
<code>Rational</code>	rational number <code>nnn/nnn</code>
<code>Complex</code>	complex number <code>nnn + nnn I</code>
<code>RawMedium</code>	raw descriptor for an input-output medium

Special objects.

Most special objects are input and output in special forms.

As an optimization for some special kinds of computations, the raw data in *Mathematica* special objects can be given explicitly using `Raw[head, "hexstring"]`. The data is specified as a string of hexadecimal digits, corresponding to an array of bytes. When no special output form exists, `InputForm` prints special objects using `Raw`. *The behaviour of Raw differs from one implementation of Mathematica to another; its general use is strongly discouraged.*